1 Objective and Introduction

◇ **Objective**

Prove $\int_{\overline{\mathcal{M}}_{1,1}} \psi = 1/24$.

Because we cannot compute this directly on the moduli space, we will instead pull the ψ class back to a family of pointed elliptic curves. By integrating the pullback, we can determine the integral over the moduli space!

◇ **The Integral Map and Chow Groups**

The integral is the pushforward of the constant map:

$$X \xrightarrow{c} \boldsymbol{pt.} \quad \Rightarrow \quad A^*(X) \xrightarrow{c_* = \int_X} A^*(\boldsymbol{pt.}).$$

Only cycles with top codimension part have non-zero integrals. In essence, the integral captures the degree of that part.

$$\begin{tikzcd}
\{\, f + tg : \ t \in \mathbb{P}^1 \,\} \arrow[r, "\pi"] \arrow[d] & \mathbb{P}^1_t \arrow[l, bend right, "p_1"'] \arrow[d, "\mu : t \mapsto [\pi^{-1}(t)]"'] \arrow[dr, "\tilde{c}"] & \\
\overline{\mathcal{U}}_{1,1} \arrow[r] & \overline{\mathcal{M}}_{1,1} \arrow[r, "c"'] \arrow[l, bend left, "\sigma"] & \boldsymbol{pt.}
\end{tikzcd}
\qquad
\begin{tikzcd}
A^*(\mathbb{P}^1_t) \arrow[dr, "\tilde{c}_* = \int_{\mathbb{P}^1}"] & \\
A^*(\overline{\mathcal{M}}_{1,1}) \arrow[u, "\mu^*"] \arrow[r, "c_* = \int_{\overline{\mathcal{M}}_{1,1}}"'] & A^*(\boldsymbol{pt.})
\end{tikzcd}$$

From here, $\int_{\overline{\mathcal{M}}_{1,1}} \psi = c_*(\psi) = \tilde{c}_*(\mu^*(\psi)) = \int_{\mathbb{P}^1} \mu^*(\psi)$.

◇ **The Psi Class**

The ψ class in $\overline{\mathcal{M}}_{1,1}$ is the first Chern class of the relative cotangent line bundle associated to the mark p_1: $\psi = c_1(\mathbb{L}_1)$.

◇ So we may outline how we're going to complete our objective:

(a) Create a family of elliptic curves.

(b) Describe the pullback of ψ in our family.

(c) Describe the classifying map into $\overline{\mathcal{M}}_{1,1}$.

(d) Integrate it and verify the result.

2 Creating a Family

Consider a family of cubics $\mathcal{E} \xrightarrow{\pi} \mathbb{P}^1_t$ given by the pencil $\{ f + tg = 0 \}$ with $t \in \mathbb{P}^1$ where

$$\begin{cases} f = x^3 - 4x + 0.01y \\ g = y^3 - 4y + 0.02x \end{cases}$$

which pass through[1] the lattice points $(-2, 2), (0, 2), (2, 2), \ldots, (2, -2)$. By picking one of them, we get a family of elliptic curves with a mark determined by the chosen point.

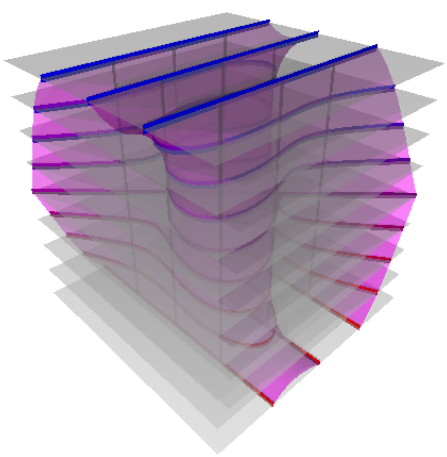

Figure 1: $\{f + tg : t \in \mathbb{P}^1\}$ with t varying in the vertical direction

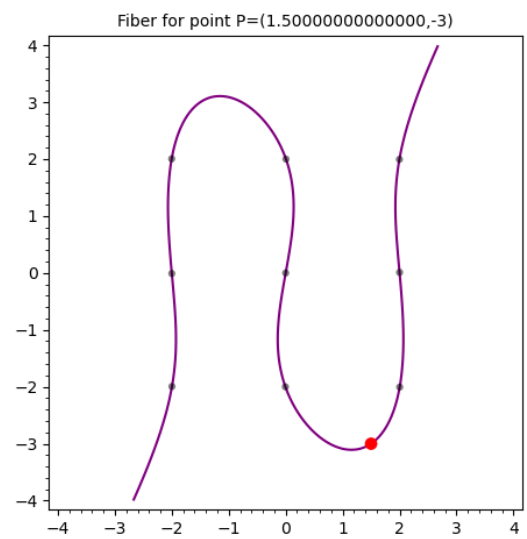

Figure 2: Typical fiber at $t = -0.177354709418838$

Exercise 1. Via Bézout's theorem, verify that two cubics in general position intersect at exactly 9 distinct points.

Exercise 2. Via the genus-degree formula $g = \binom{d-1}{2}$, verify that all smooth curves in our family have genus one (and thus, are elliptic).

Exercise 3. Convince yourself that choosing one of the intersection points p_i defines a section $p_i : \mathbb{P}^1_t \to \mathcal{E}$, framing this as a family of pointed elliptic curves.

Exercise 4. Draw the tangent to the fiber at $(0, 2)$. Transport that drawing to the family. Convince yourself that the relative tangent bundle to the curves at the section is the same as the normal bundle of the section.

[1] For numerical reasons, they don't pass exactly through these points but very close.

3 Describing the Classifying Map

Our family induces a classifying map

$$\mu : \mathbb{P}^1 \to \overline{\mathcal{M}}_{1,1}, \quad t \mapsto [\pi^{-1}(t)],$$

which maps smooth/singular curves to smooth/singular points in the moduli space[2]. The locus of singular curves forms an irreducible boundary divisor $\delta_{\text{irr}} \subset \overline{\mathcal{M}}_{1,1}$. Since all singular fibers map to this divisor, we can read the degree of μ simply by counting them. We count them as follows.

Exercise 5. The total space $\mathcal{E}$ is the blowup of $\mathbb{P}^2$ at the 9 common points. Calculate its Euler characteristic $\chi(\mathcal{E})$ via *additivity over disjoint unions*:

$$\chi(\mathcal{E}) = \chi(\mathbb{P}^2) - 9 \cdot \chi(\boldsymbol{pt.}) + 9 \cdot \chi(\mathbb{P}^1) = _____ - 9_____ + 9_____ = _____.$$

Exercise 6. Find the Euler characteristic of a nodal cubic:

$$\chi(\text{nodal}) = \chi(\mathbb{P}^1) - 2\chi(\boldsymbol{pt.}) + \chi(\boldsymbol{pt.}) = _____ - 2_____ + _____ = _____.$$

Exercise 7. Via the *multiplicativity over fibrations* for the Euler characteristic, $\chi(\mathcal{E}) = \chi(F)\chi(B)$, realize that:

$$\chi(\mathcal{E}) = (\text{number of singular fibers}) \cdot \chi(\text{nodal}).$$

From this, find the degree of the map μ via the previous exercise.

4 Final Computation

To compute the final integral, we use everything we've gathered up to now. Explicitly, we need:

$$\psi_1 = -c_1\left(\mathcal{O}_E(E)\right) = -(-\boldsymbol{pt.}).$$

and the fact that the map from the coarse moduli space $\overline{M}_{1,1}$ to the stack $\overline{\mathcal{M}}_{1,1}$ has degree 2.

Exercise 8. Fill in the blanks to complete the proof:

$$\int_{\overline{\mathcal{M}}_{1,1}} \psi = \frac{1}{2\deg(\mu)} \int_{\mathbb{P}^1} \mu^*\psi = \frac{1}{2 \cdot (_____)} \int_{\mathbb{P}^1} _____ = _____.$$

[2]Strictly speaking, this is the induced map at the level of families.